Cell Physiology: Functions of different components

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OBJECTIVES

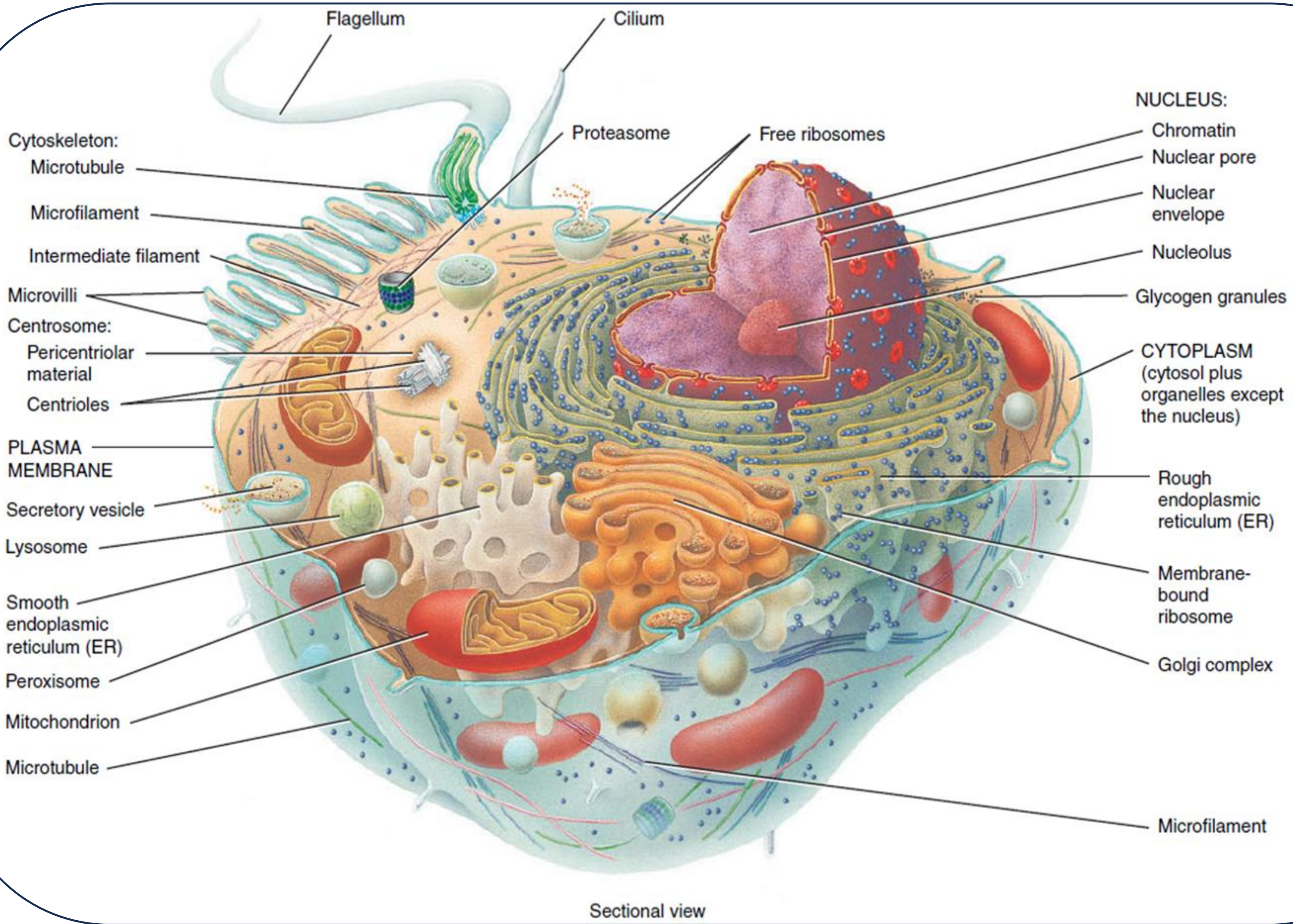
By the end of this lecture students will be able to:

- Name and describe main parts of a cell.
- Describe the structure and function of cytoplasm, cytosol and organelles.
- Describe the structure and function of the nucleus.
- Describe how cells differ in size and shape.
- Describe the cellular changes that occur with aging.

The Cellular Level of Organization

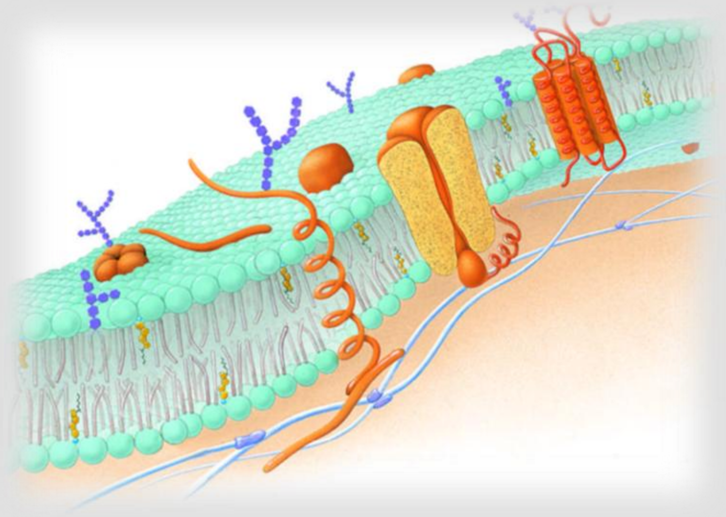
- The **cell** is the basic living, structural, and functional unit of the body.
- The average adult human body consists of more than 100 trillion cells.
- All cells arise from existing cells by the process of **cell division**, in which one cell divides into two identical cells.
- Different types of cells fulfill unique roles that support homeostasis and contribute to the many functional capabilities of the human organism.
- **Cell biology** or cytology is the study of cellular structure and function.
- Cell structure and function are intimately related.
- Cells carry out a dazzling array of chemical reactions to create and maintain life processes.
- Cells are divided into three main parts: **plasma membrane**, **cytoplasm** **and** **nucleus**.

Parts of a Cell



1. The **plasma membrane**

- forms the cell's flexible outer surface,
- separating the cell's *internal environment* from the *external environment.*
- It is a selective barrier that regulates the flow of materials into and out of a cell.
- This selectivity helps establish and maintain the appropriate environment for normal cellular activities.
- The plasma membrane also plays a key role in communication among cells and between cells and their external environment.



2. The **cytoplasm**

- consists of all the cellular contents between the plasma membrane and the nucleus.
- This compartment has two components: cytosol and organelles.
- **Cytosol**, the fluid portion of cytoplasm, also called ***intracellular fluid***, contains water, dissolved solutes, and suspended particles.
- Within the cytosol are several different types of ***organelles,*** each type of organelle has a characteristic shape and specific functions. Examples include the cytoskeleton, ribosomes, endoplasmic reticulum, Golgi complex, lysosomes, peroxisomes, and mitochondria.

Cytosol

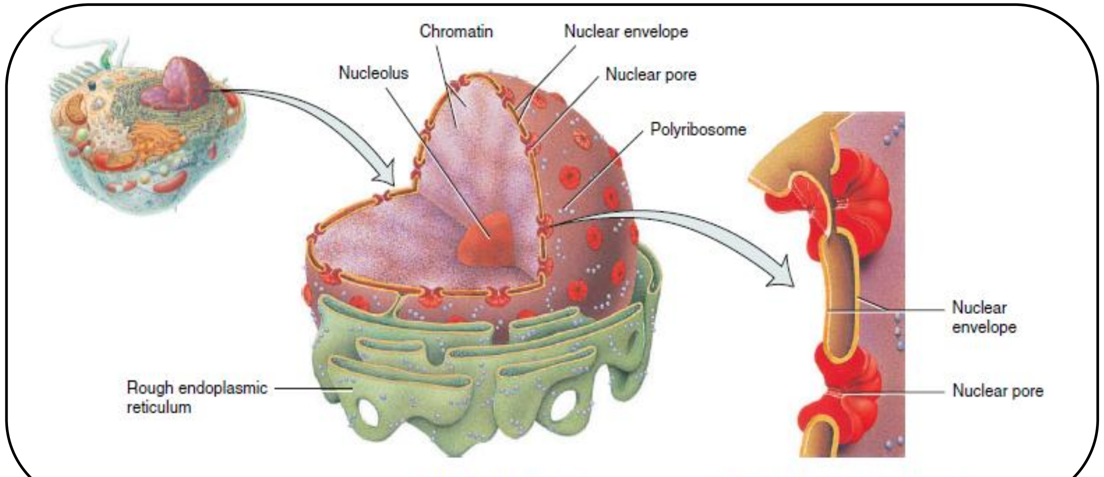
- The **cytosol (***intracellular fluid***)** is the fluid portion of the cytoplasm that surrounds organelles and constitutes about 55% of total cell volume.
- Consists of 75–90% water plus various dissolved and suspended components like ions, glucose, amino acids, fatty acids, proteins, lipids, ATP, and waste products, and organic molecules that aggregate into masses for storage. These aggregations may appear and disappear at different times in the life of a cell.
- Examples include *lipid droplets* that contain triglycerides, and clusters of glycogen molecules called **glycogen granules**.
- The cytosol is the site of many chemical reactions required for a cell's existence.   Eg: in cytosol catalyze *glycolysis,* a series of 10 chemical reactions that produce two molecules of ATP from one molecule of glucose.
- Other types of cytosolic reactions provide the building blocks for maintenance of cell structures and for cell growth.

3. The **nucleus**

- The **nucleus** is a spherical or oval-shaped structure that usually is the most prominent feature of a cell. houses most of a cell's DNA
- Within the nucleus, each **chromosome** a single molecule of DNA associated with several proteins, contains thousands of hereditary units called **genes** that control most aspects of cellular structure and function.
- Most cells have a single nucleus, although some, such as mature red blood cells, have none. In contrast, skeletal muscle cells and a few other types of cells have multiple nuclei.

FUNCTIONS OF NUCLEUS

1. Controls cellular structure.
2. Directs cellular activities.
3. Produces ribosomes in nucleoli.



(a) Details of the nucleus

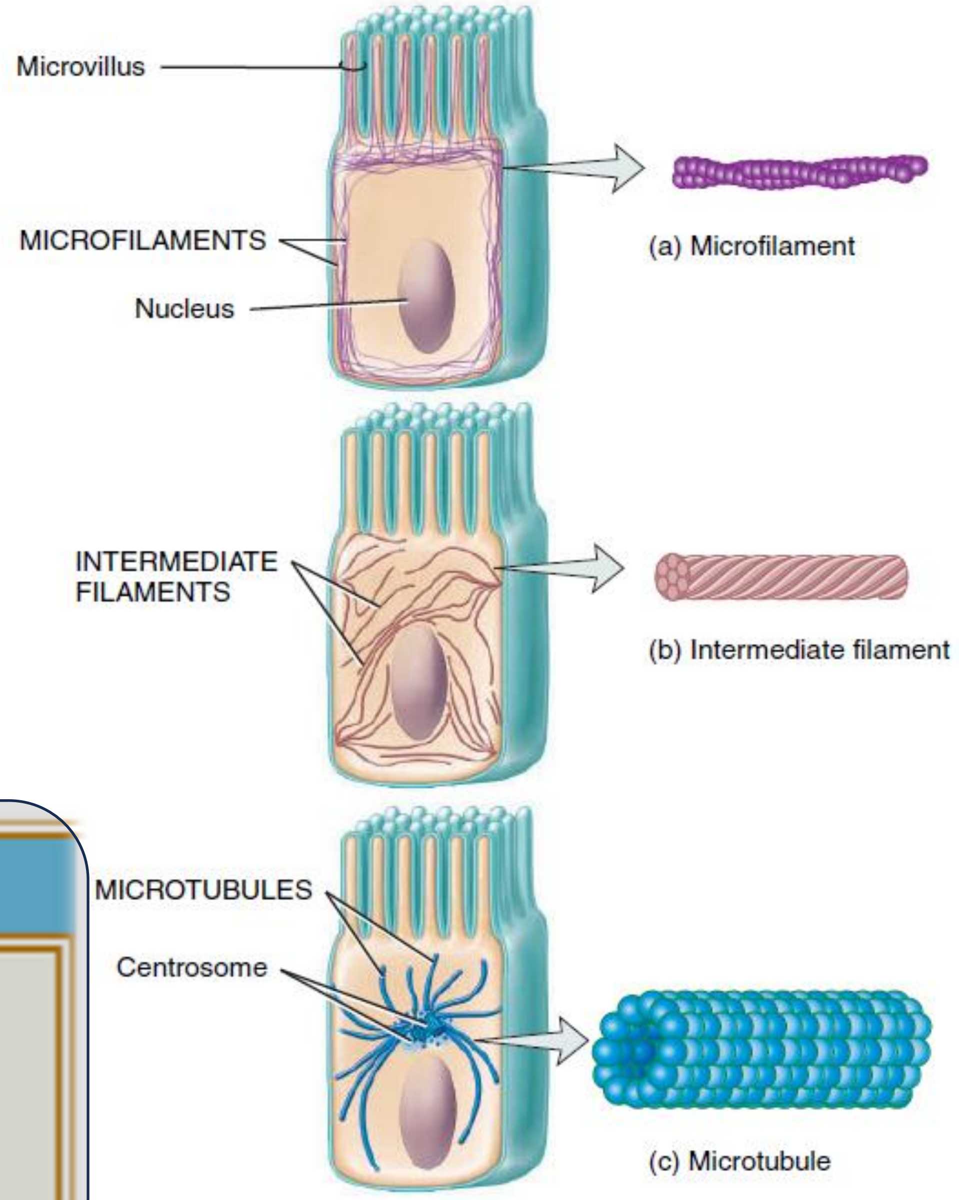
(b) Details of the nuclear envelope

3. The **nucleus (cont…)**

- A double membrane called the **nuclear envelope** separates the nucleus from the cytoplasm.
- Both layers of the nuclear envelope are lipid bilayers similar to the plasma membrane. The outer membrane of the nuclear envelope is continuous with rough ER and resembles it in structure.
- Many openings called **nuclear pores** extend through the nuclear envelope, which control the movement of substances between the nucleus and cytoplasm.
- **Nucleoli**, which produce ribosomes; and genes arranged on chromosomes, which control cellular structure and direct cellular activities.
- Human somatic cells have 46 chromosomes, 23 inherited from each parent. The total genetic information carried in a cell or an organism is its genome.

The **cytoskeleton**

- is a network of protein filaments that extends throughout the cytosol.
- Three types of filaments contribute to the cytoskeleton’s structure, as well as the structure of other organelles. In the order of their increasing diameter, these structures are **microfilaments, intermediate filaments, and microtubules.**



FUNCTIONS OF THE CYTOSKELETON

1. Serves as a scaffold that helps determine a cell’s shape and organize the cellular contents.
2. Aids movement of organelles within the cell, of chromosomes during cell division, and of whole cells such as phagocytes.

Organelles

- **Organelles** are specialized structures within the cell that have characteristic shapes, and they perform specific functions in cellular growth, maintenance, and reproduction.
- Despite the many chemical reactions going on in a cell at any given time, there is little interference among reactions because they are confined to different organelles.
- Each type of organelle has its own set of enzymes that carry out specific reactions, and serves as a functional compartment for specific biochemical processes.
- The numbers and types of organelles vary in different cells, depending on the cell's function.
- Although they have different functions, organelles often cooperate to maintain homeostasis.

Mitochondria:

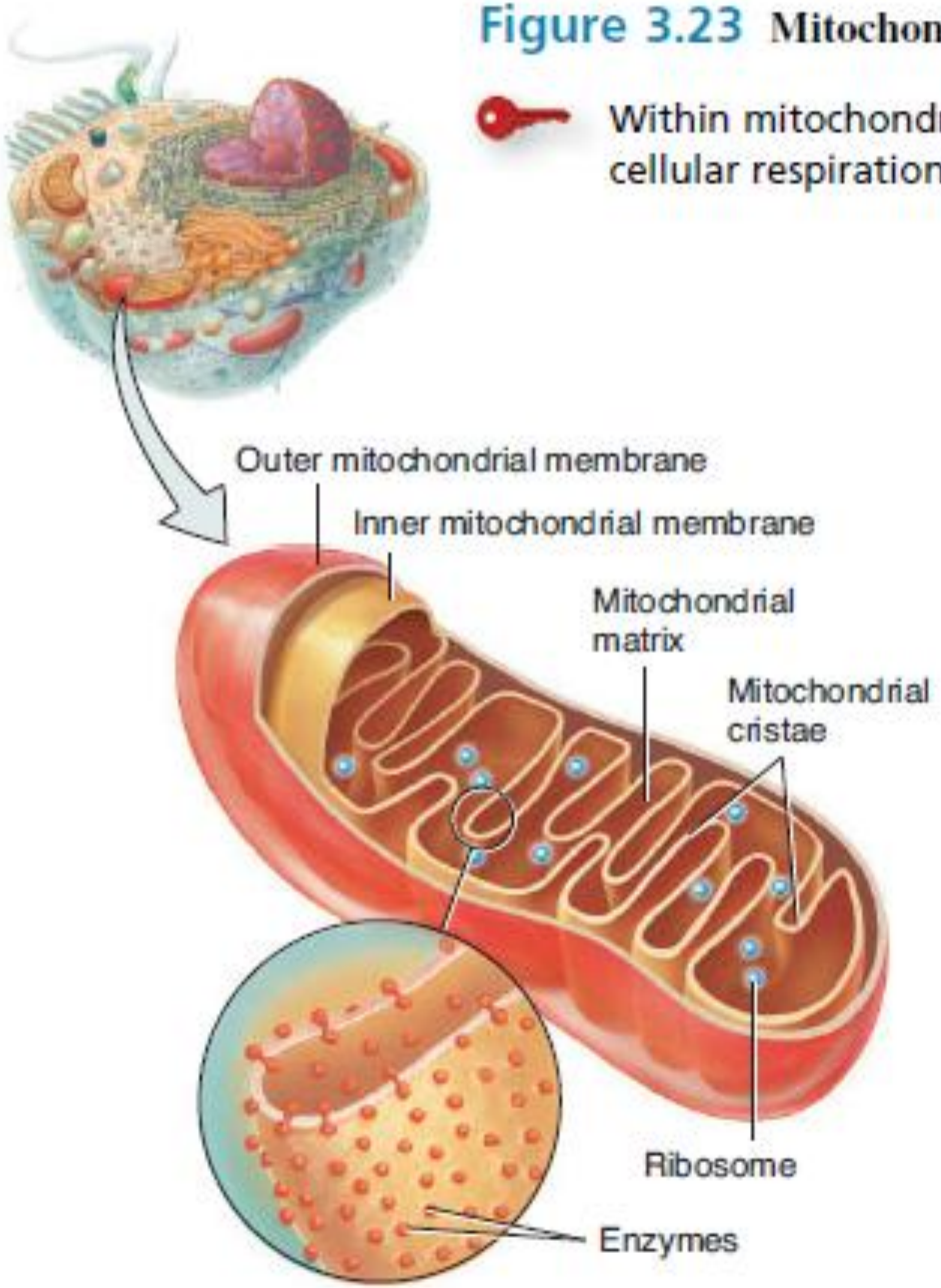
- **mitochondria** (singular is *mitochondrion*) are the “powerhouses” of the cell Because they generate most of the ATP through aerobic (oxygen requiring) respiration,.
- A cell may have as few as a hundred or as many as several thousand mitochondria, depending on its activity. Active cells that use ATP at a high rate such as those found in the muscles, liver, and kidneys have a large number of mitochondria. For example, regular exercise can lead to an increase in the number of mitochondria in muscle cells, which allows muscle cells to function more efficiently.
- Mitochondria are usually located within the cell where oxygen enters the cell or where the ATP is used, for example, among the contractile proteins in muscle cells.
- Mitochondria also play an important and early role in **apoptosis, (**the orderly, genetically programmed death of a cell).

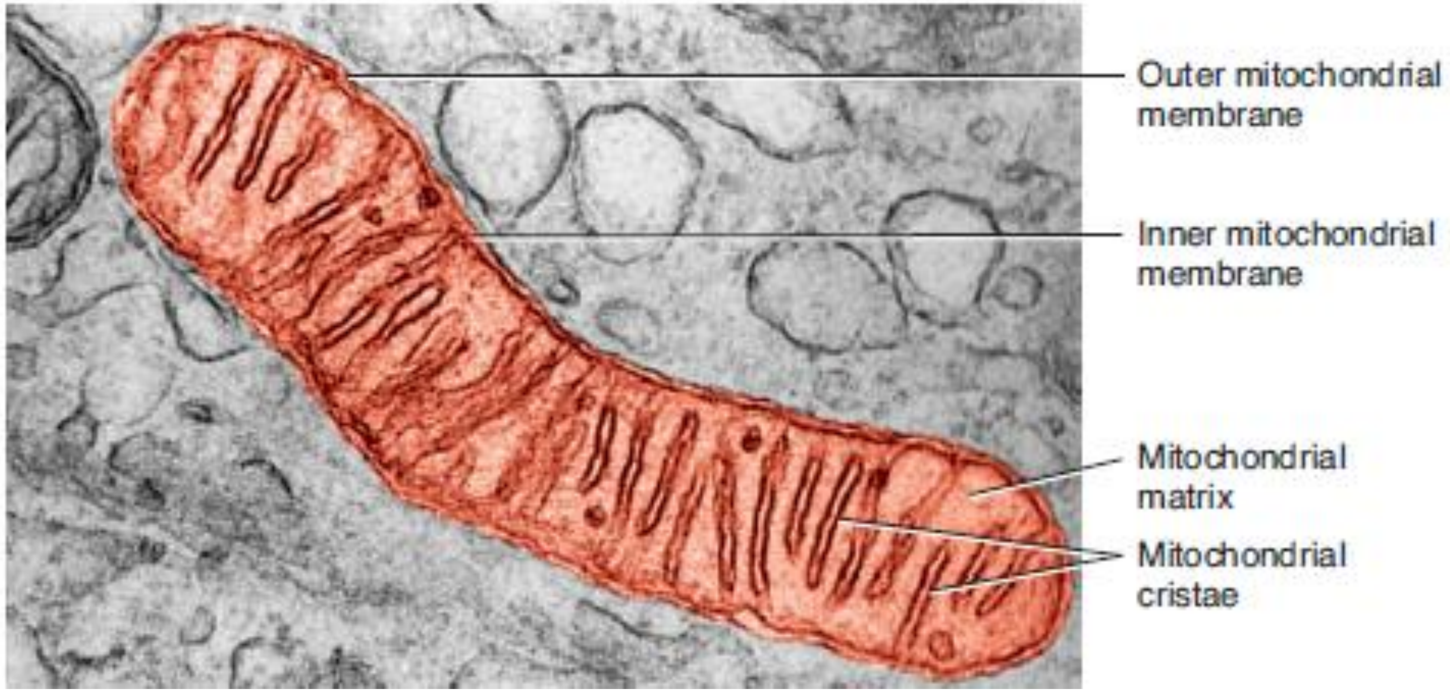
Figure 3.23 **Mitochondria.**

Within mitochondria, chemical reactions of aerobic cellular respiration generate ATP.

FUNCTIONS OF MITOCHONDRIA

1. Generate ATP through reactions of aerobic cellular respiration.
2. Play an important early role in apoptosis.



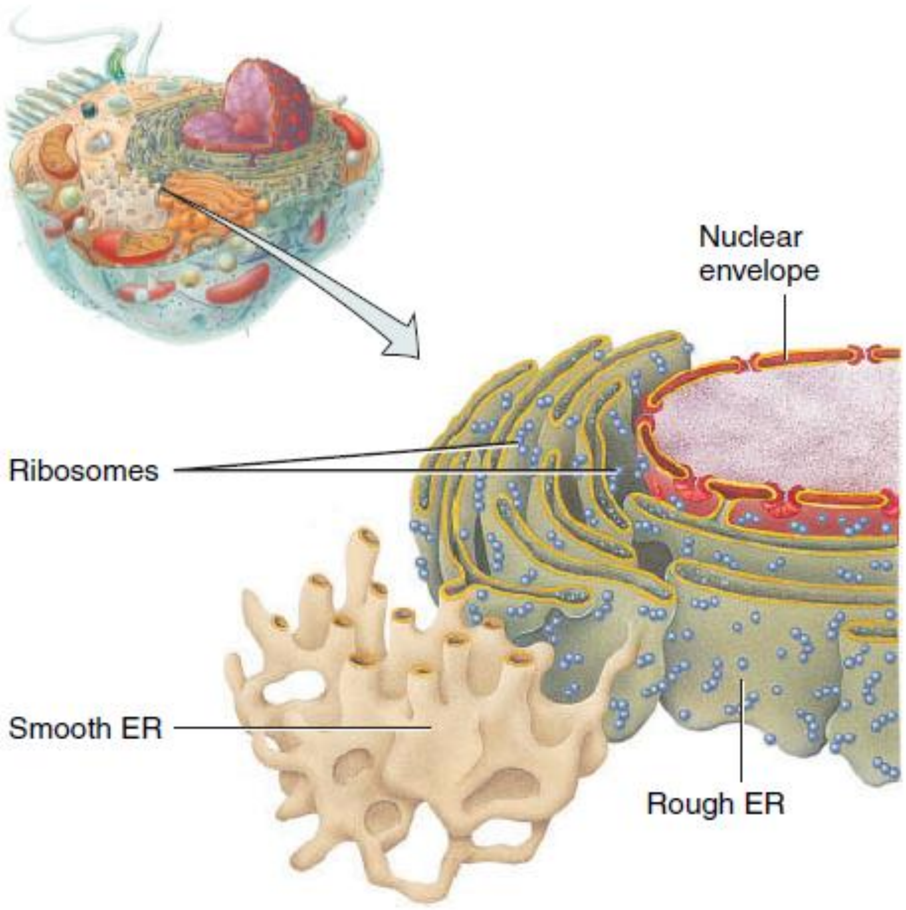


Endoplasmic Reticulum:

- The **endoplasmic reticulum (ER)** (*plasmic* = cytoplasm; *reticulum* =network) is a network of membranes in the form of flattened sacs or tubules.
- The ER extends from the nuclear envelope (membrane around the nucleus), to which it is connected and projects throughout the cytoplasm.
- The ER is so extensive that it constitutes more than half of the membranous surfaces within the cytoplasm of most cells.
- Cells contain two distinct forms of ER, which differ in structure and function. (**Rough ER** and **Smooth ER)**

Figure 3.19 Endoplasmic reticulum.

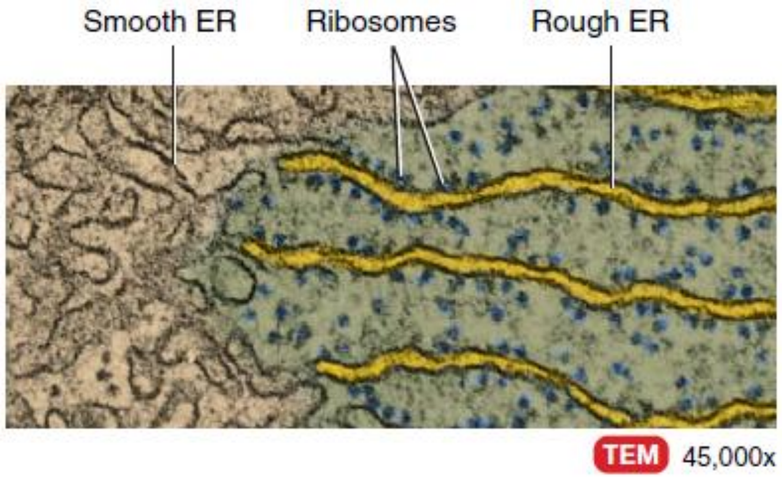
The endoplasmic reticulum is a network of membrane-enclosed sacs or tubules that extend throughout the cytoplasm and connect to the nuclear envelope.



(a) Details

FUNCTIONS OF ENDOPLASMIC RETICULUM

1. Rough ER synthesizes glycoproteins and phospholipids that are transferred into cellular organelles, inserted into the plasma membrane, or secreted during exocytosis.
2. Smooth ER synthesizes fatty acids and steroids, such as estrogens and testosterone; inactivates or detoxifies drugs and other potentially harmful substances; removes the phosphate group from glucose-6-phosphate; and stores and releases calcium ions that trigger contraction in muscle cells.



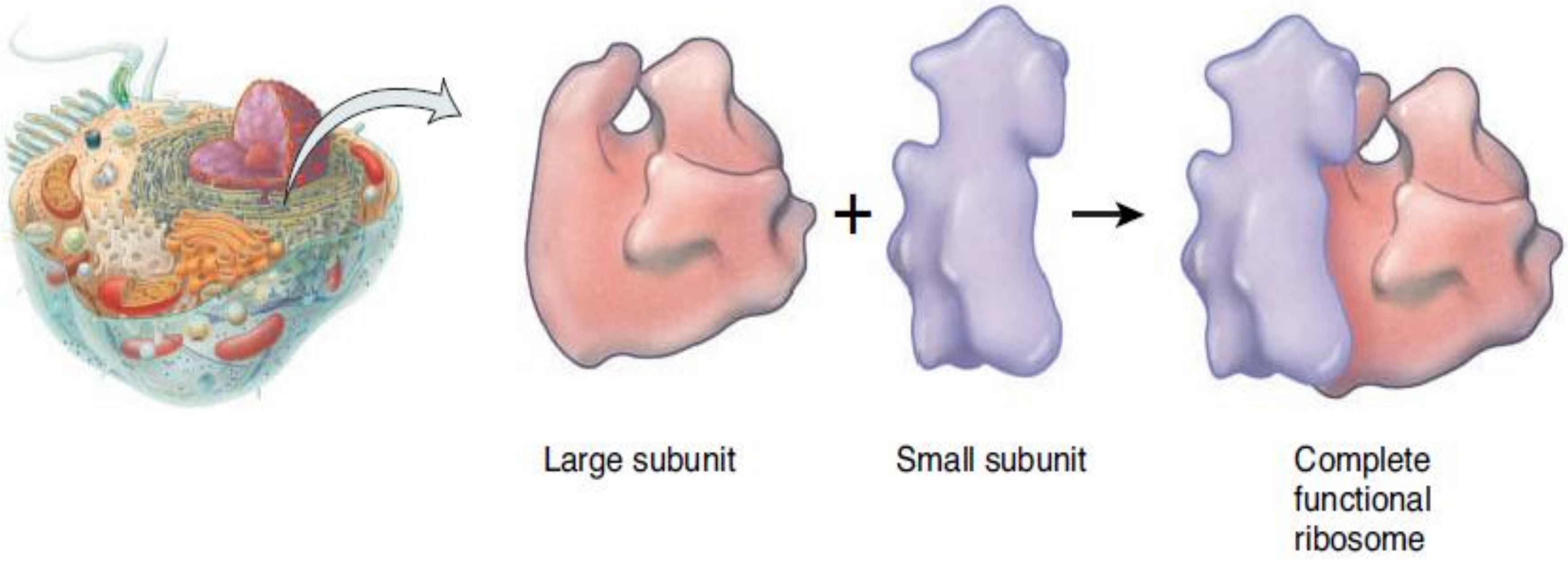
(b) Transverse section

Figure 3.18 Ribosomes.

Ribosomes are the sites of protein synthesis.

FUNCTIONS OF RIBOSOMES

1. Ribosomes associated with endoplasmic reticulum synthesize proteins destined for insertion in the plasma membrane or secretion from the cell.
2. Free ribosomes synthesize proteins used in the cytosol.

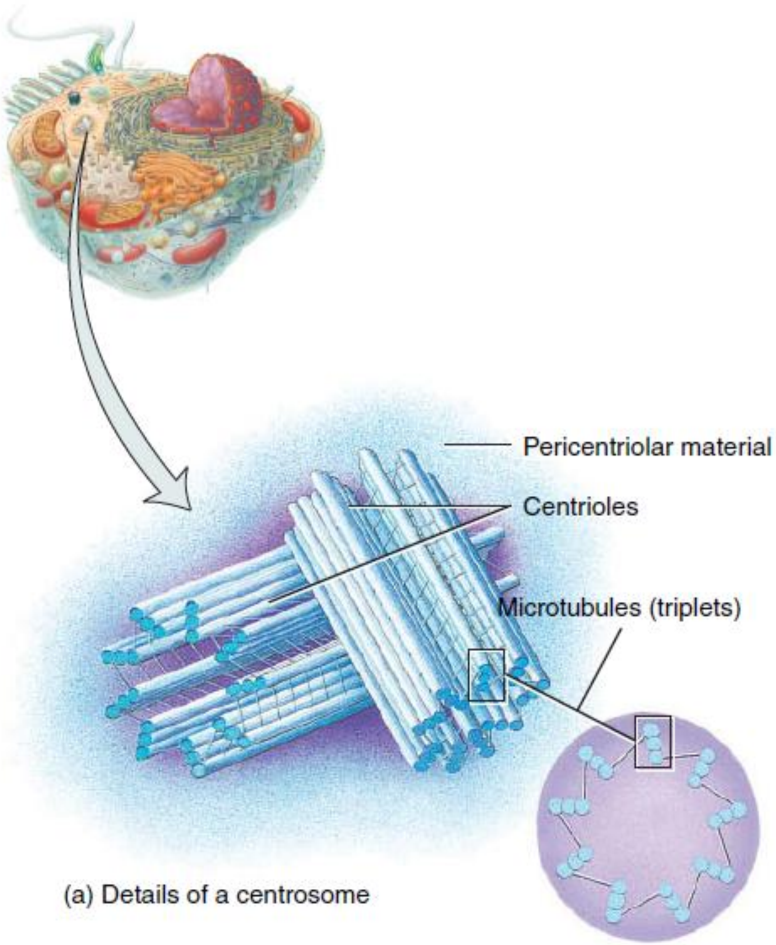


Details of ribosomal subunits

Centrosome

Figure 3.16 **Centrosome.**

Located near the nucleus, the centrosome consists of a pair of centrioles and pericentriolar material.

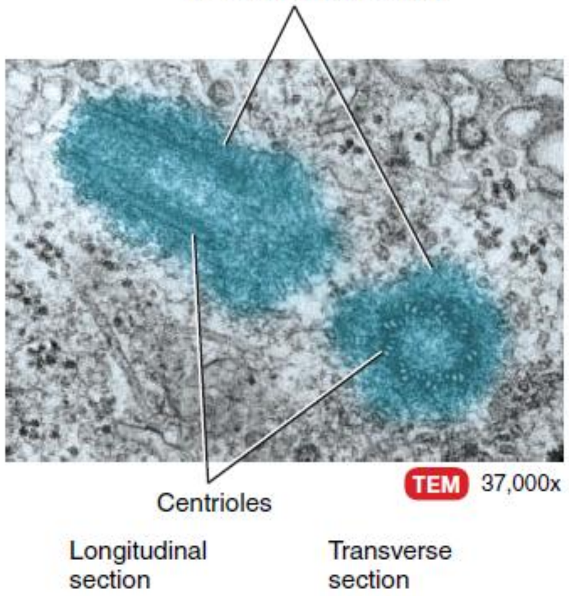


(a) Details of a centrosome

(b) Arrangement of microtubules in centrosome

FUNCTIONS OF THE CENTROSOMES

1. The pericentriolar material of the centrosome contains tubulins that build microtubules in nondividing cells.
2. The pericentriolar material of the centrosome forms the mitotic spindle during cell division.



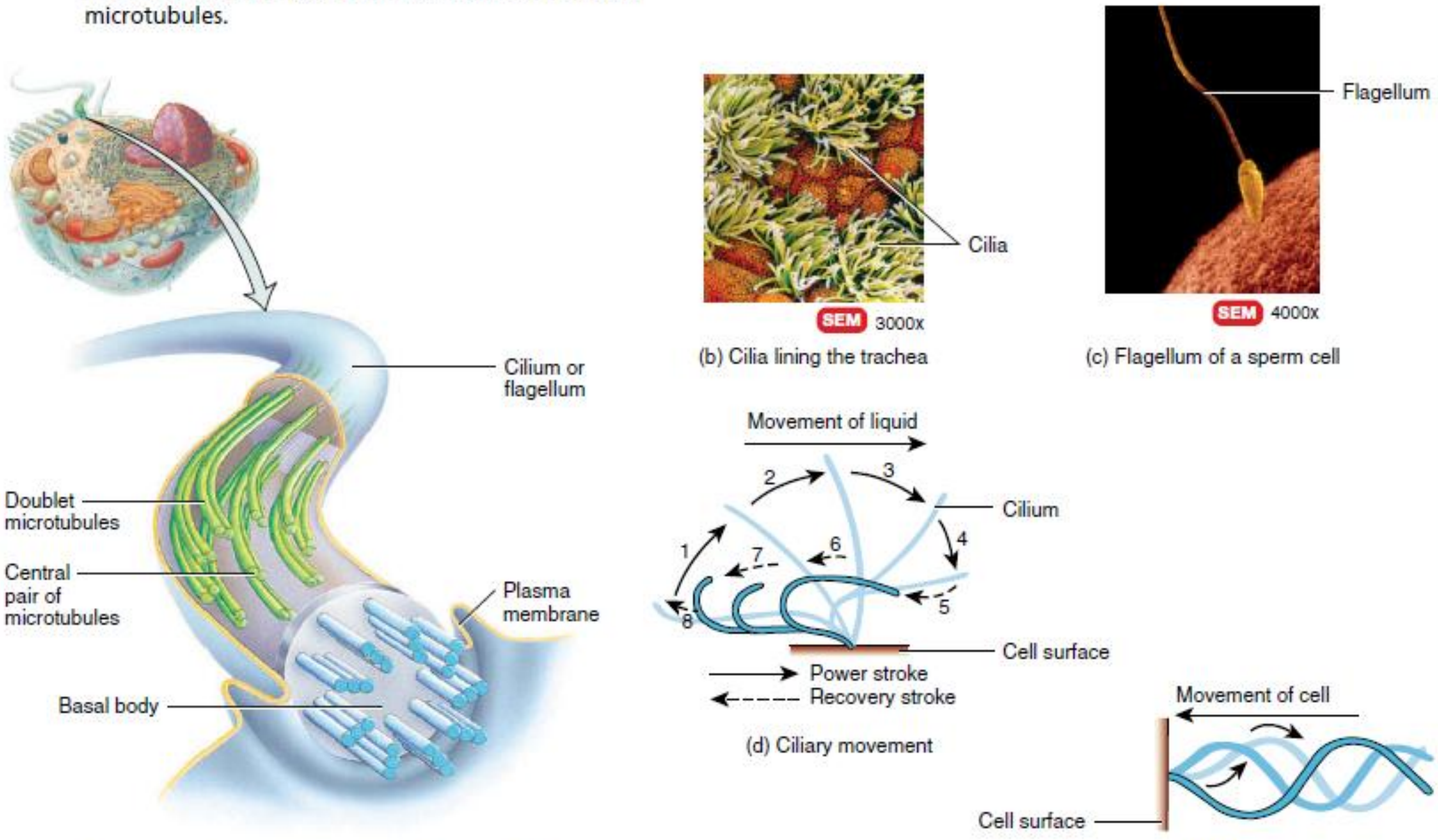
(c) Centrioles

Figure 3.17 Cilia and flagella.

A cilium contains a core of microtubules with one pair in the center surrounded by nine clusters of doublet microtubules.

FUNCTIONS OF CILIA AND FLAGELLA

1. Cilia move fluids along a cell's surface.
2. A flagellum moves an entire cell.



(a) Arrangement of microtubules in a cilium or flagellum

(b) Cilia lining the trachea

(c) Flagellum of a sperm cell

(d) Ciliary movement

(e) Flagellar movement

CLINICAL CONNECTION | *Cilia and Smoking*

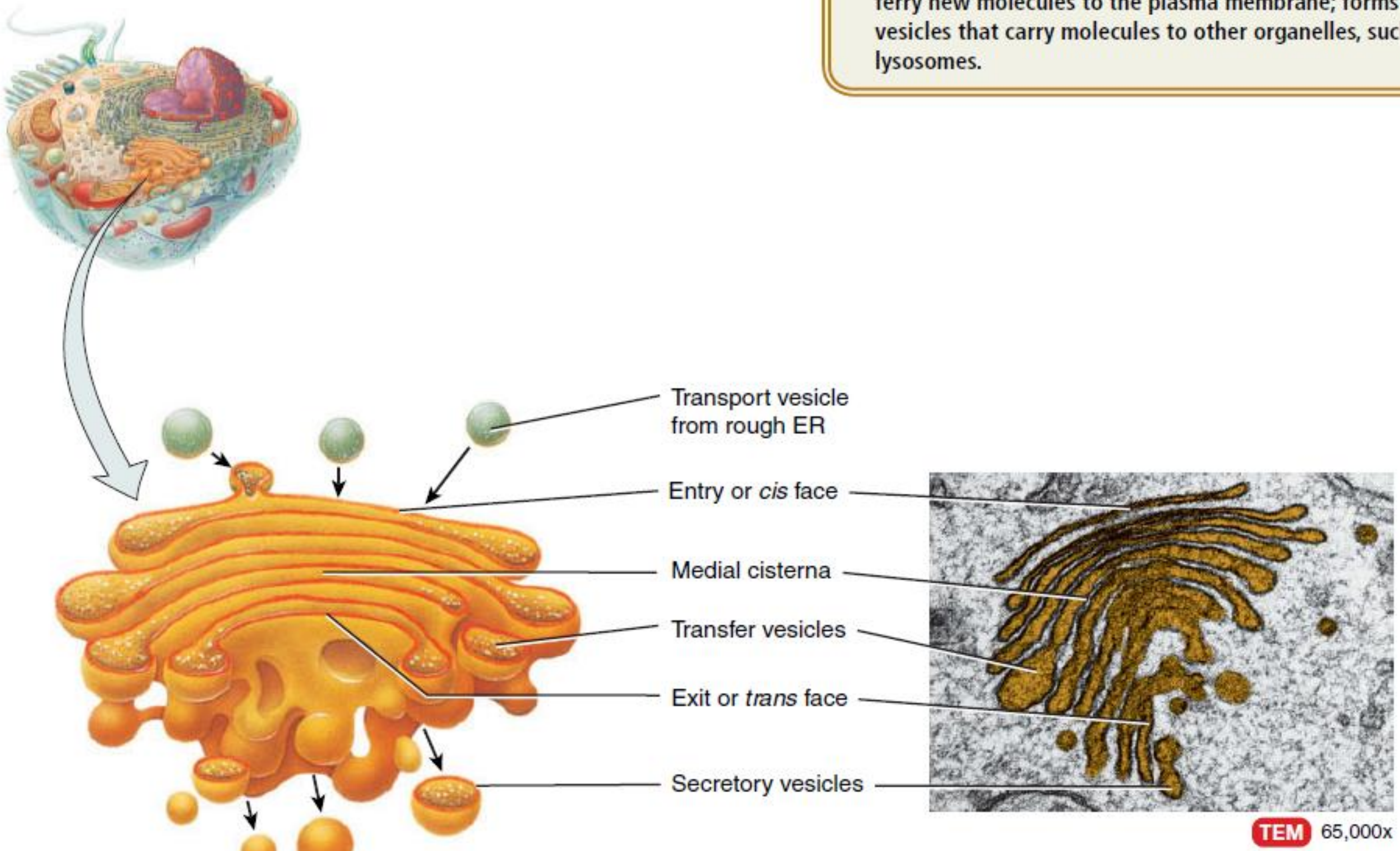
The movement of cilia is paralyzed by nicotine in cigarette smoke. For this reason, smokers cough often to remove foreign particles from their airways. Cells that line the uterine (fallopian) tubes also have cilia that sweep oocytes (egg cells) toward the uterus, and females who smoke have an increased risk of ectopic (outside the uterus) pregnancy. •

Figure 3.20 Golgi complex.

The opposite faces of a Golgi complex differ in size, shape, content, and enzymatic activities.

FUNCTIONS OF THE GOLGI COMPLEX

1. Modifies, sorts, packages, and transports proteins received from the rough ER.
2. Forms secretory vesicles that discharge processed proteins via exocytosis into extracellular fluid; forms membrane vesicles that ferry new molecules to the plasma membrane; forms transport vesicles that carry molecules to other organelles, such as lysosomes.

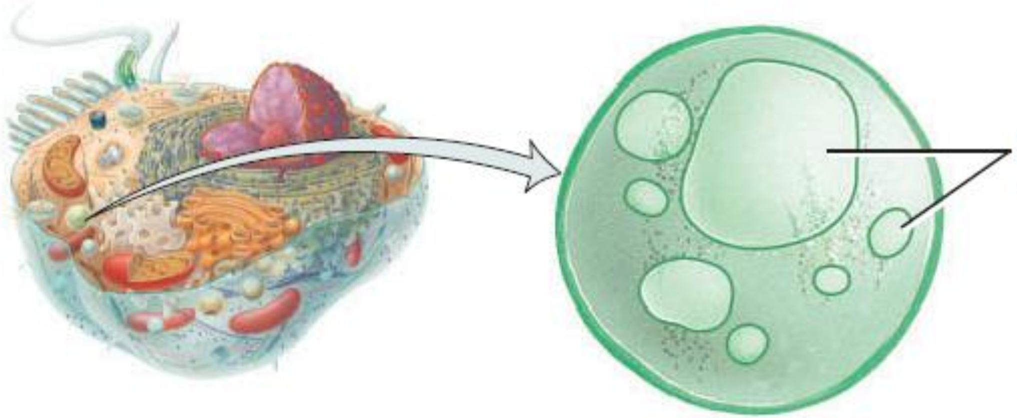


(a) Details

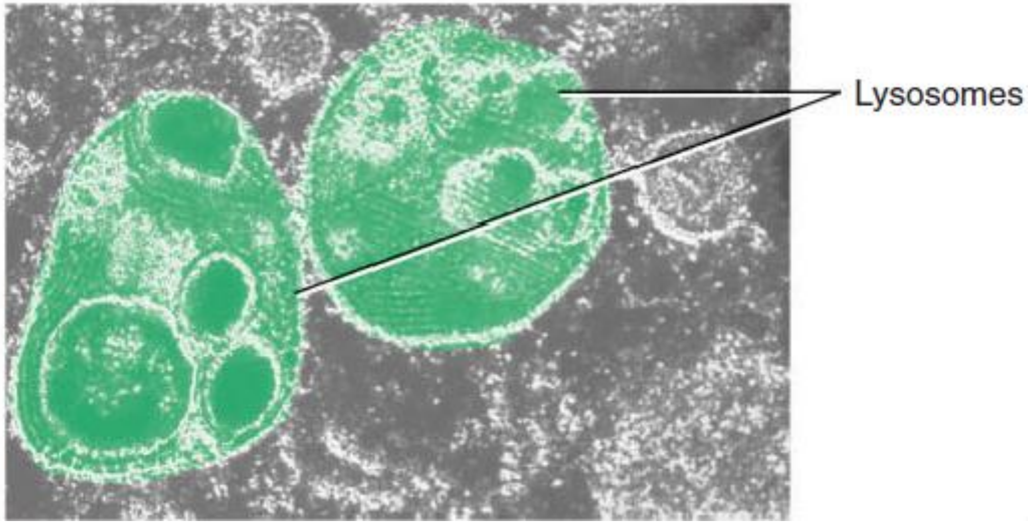
(b) Transverse section

Figure 3.22 Lysosomes.

Lysosomes contain several types of powerful digestive enzymes.



(a) Lysosome



FUNCTIONS OF LYSOSOMES

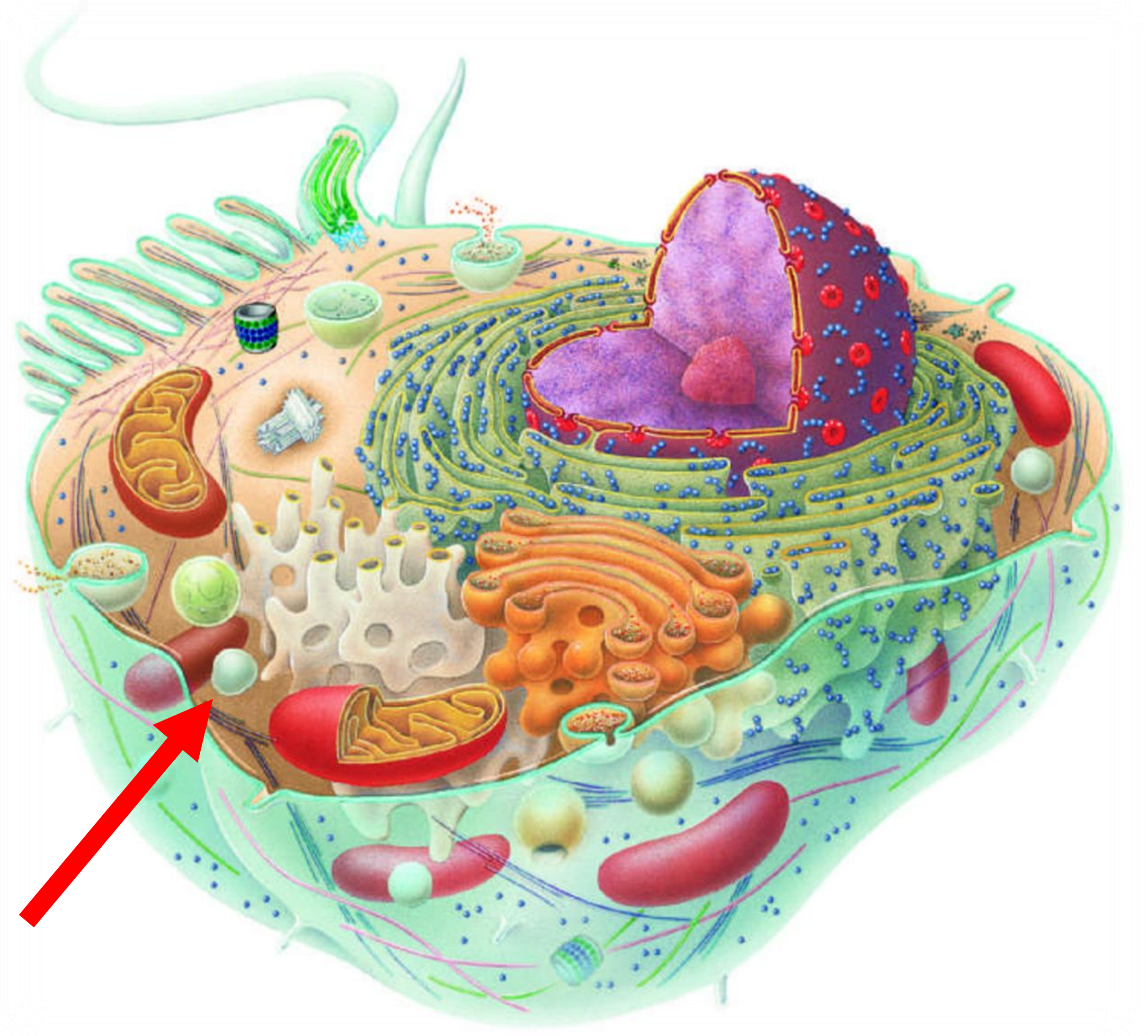
1. Digest substances that enter a cell via endocytosis and transport final products of digestion into cytosol.
2. Carry out autophagy, the digestion of worn-out organelles.
3. Implement autolysis, the digestion of an entire cell.
4. Accomplish extracellular digestion.

CLINICAL CONNECTION | *Tay-Sachs Disease*

Some disorders are caused by faulty or absent lysosomal enzymes. For instance, **Tay-Sachs disease** (TĀ-SAKS), which most often affects children of Ashkenazi (eastern European Jewish) descent, is an inherited condition characterized by the absence of a single lysosomal enzyme called Hex A. This enzyme normally breaks down a membrane glycolipid called ganglioside G_{M2} that is especially prevalent in nerve cells. As the excess ganglioside G_{M2} accumulates, the nerve cells function less efficiently. Children with Tay-Sachs disease typically experience seizures and muscle rigidity. They gradually become blind, demented, and uncoordinated and usually die before the age of 5. Tests can now reveal whether an adult is a carrier of the defective gene. •

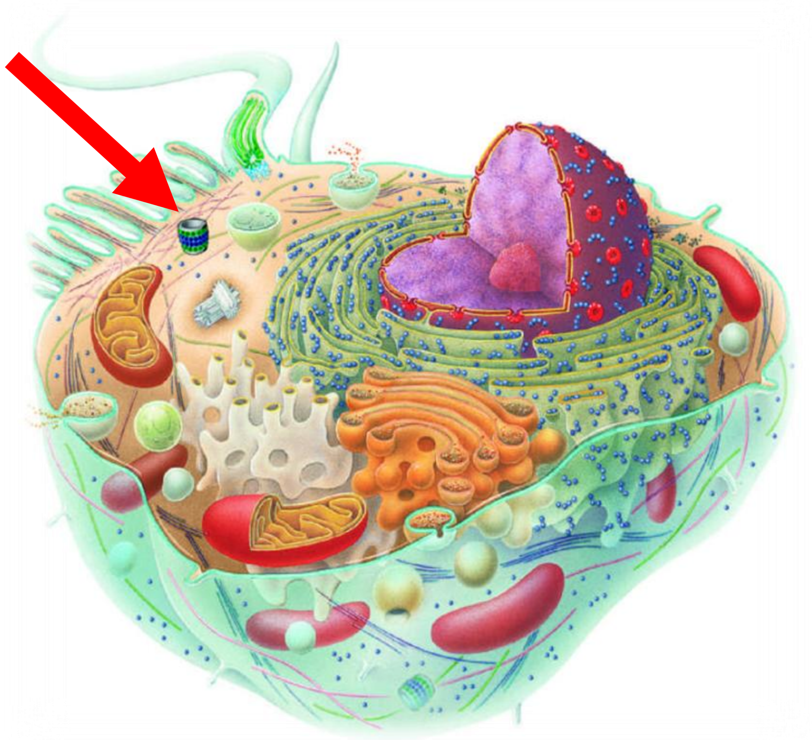
Peroxisomes

- Another group of organelles similar in structure to lysosomes, but smaller, are the **peroxisomes**.
- Peroxisomes, also called *microbodies,* contain several *oxidases,* enzymes that can oxidize (remove hydrogen atoms from) various organic substances. For instance, amino acids and fatty acids are oxidized in peroxisomes as part of normal metabolism. In addition, enzymes in peroxisomes oxidize toxic substances, such as alcohol. Thus, peroxisomes are very abundant in the liver, where detoxification of alcohol and other damaging substances occurs.



Proteasomes

Continuous destruction of unneeded, or faulty proteins is the function of tiny barrel-shaped structures consisting of four stacked rings of proteins around a central core called **proteasomes** (protein bodies). For example, proteins that are part of metabolic pathways need to be degraded after they have accomplished their function.

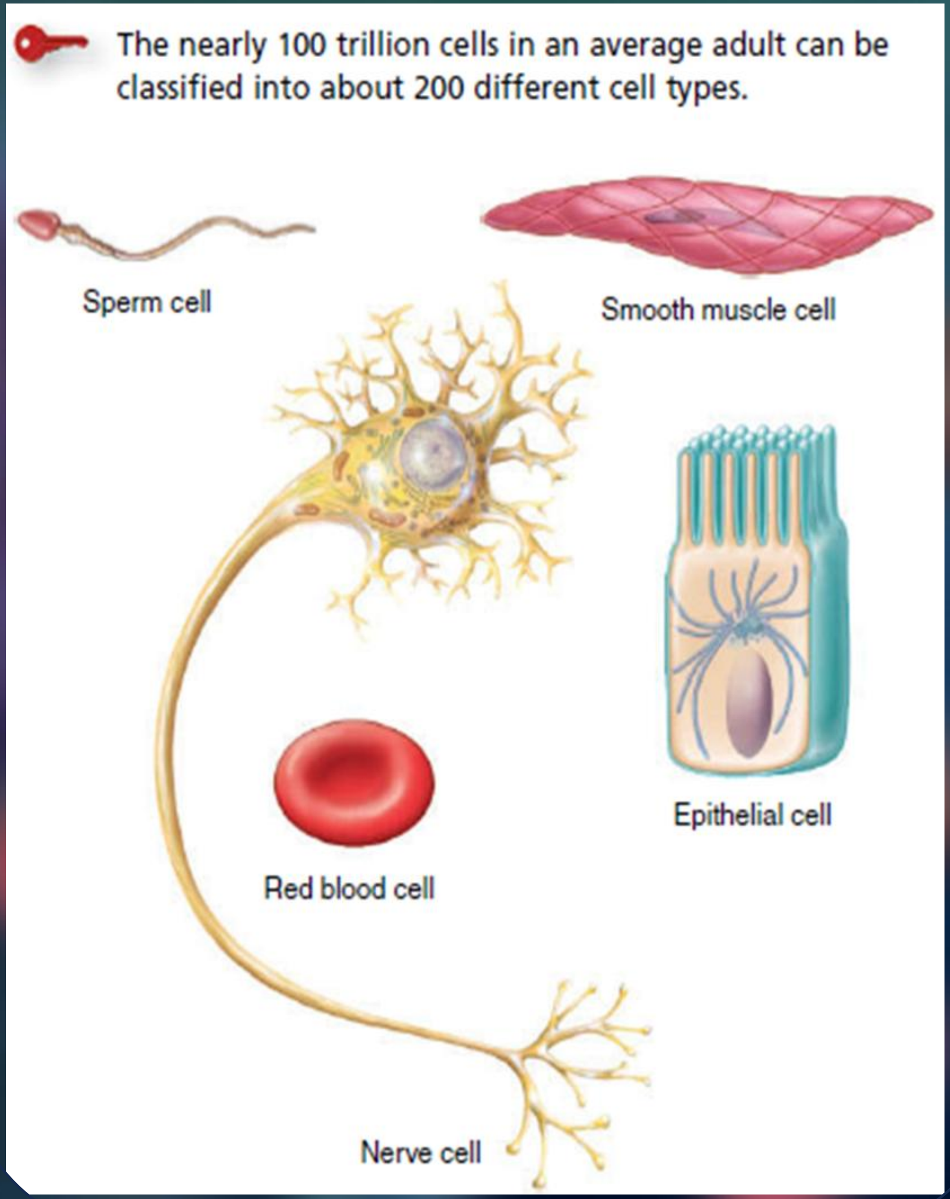


CLINICAL CONNECTION | *Proteasomes and Disease*

Some diseases could result from failure of proteasomes to degrade abnormal proteins. For example, clumps of misfolded proteins accumulate in brain cells of people with Parkinson's disease and Alzheimer's disease. Discovering why the proteasomes fail to clear these abnormal proteins is a goal of ongoing research.

Cellular Diversity

1. The almost 200 different types of cells in the body vary considerably in size and shape.
2. The sizes of cells are measured in micrometers. Cells in the body range from 8 micrometers to 140 micrometers in size.
3. A cell’s shape is related to its function.



Aging and Cells

- Aging is a normal process accompanied by progressive alteration of the body's homeostatic adaptive responses.
- Many theories of aging have been proposed, including genetically programmed cessation of cell division, buildup of free radicals, and an intensified autoimmune response.

CLINICAL CONNECTION | *Free Radicals*

Free radicals produce oxidative damage in lipids, proteins, or nucleic acids by "stealing" an electron to accompany their unpaired electrons. Some effects are wrinkled skin, stiff joints, and hardened arteries. Normal metabolism—for example, aerobic cellular respiration in mitochondria—produces some free radicals. Others are present in air pollution, radiation, and certain foods we eat. Naturally occurring enzymes in peroxisomes and in the cytosol normally dispose of free radicals. Certain dietary substances, such as vitamin E, vitamin C, beta-carotene, zinc, and selenium, are referred to as antioxidants because they inhibit the formation of free radicals.

THANK YOU